Lecture 6

Flows, symmetries and the symplectic form

Objectives. *Lecture 5 made the observables an algebra. A time-independent observable changes along a motion at the rate of its bracket with H. This lecture asks what flow an observable generates, and what that flow preserves. Every observable f generates a flow, as H generates the motion. Momentum generates translations, and angular momentum generates rotations. The Jacobi identity ensures that the flows preserve the bracket. Brackets of observables determine Lie brackets of their vector fields. Noether's theorem now works in both directions: a quantity is conserved exactly when H is invariant under its flow. The bracket also determines a bilinear form ω, a sum of oriented areas. The flows preserve this form, so their Jacobian matrices are symplectic. Quadratic Hamiltonians give the Lie algebra $\mathfrak{sp}(2n)$. These results prepare the course's second answer in Lecture 7.*

By the end of this lecture you will be able to:

- Compute the Hamiltonian vector field generated by an observable.
- Relate bracket preservation by Hamiltonian flows to the Jacobi identity.
- Test whether a linear map preserves the symplectic form.

6.1 Hamiltonian vector fields: $X_f = J\nabla f$

As in Lecture 5, $\mathcal{M} \subseteq \mathbb{R}^{2n}$ is open, with coordinates $x = (q, p)$, and $J = \left(\begin{smallmatrix} 0 & I \\ -I & 0 \end{smallmatrix}\right)$ is the matrix of (4.9). Observables are C^∞ functions on $\mathcal{M}$ (Definition 5.1); in this lecture they do not depend on t.

Definition 6.1 (Hamiltonian vector field)**.** The *Hamiltonian vector field* of $f \in C^\infty(\mathcal{M})$ is

$$X_f = J\nabla f = \Big(\frac{\partial f}{\partial p}, -\frac{\partial f}{\partial q}\Big).$$

Its flow φ_s^f (Theorem 1.6) is the *flow generated by* f; the parameter s need not be a time.

A vector field F on $\mathcal{M}$ acts on functions by differentiation along it, $Fh = \nabla h \cdot F$. Along an integral curve γ, $\frac{\mathrm{d}}{\mathrm{d}s} h(\gamma(s)) = (Fh)(\gamma(s))$. For X_f this is a bracket:

$$X_f h = \nabla h \cdot J\nabla f = \sum_{i=1}^{n} \Big(\frac{\partial h}{\partial q_i}\frac{\partial f}{\partial p_i} - \frac{\partial h}{\partial p_i}\frac{\partial f}{\partial q_i} \Big) = \{h, f\}\,. \tag{6.1}$$

So along the flow of f, every observable h changes at the rate $\{h, f\}$, exactly as (5.2) says with f in place of H; Hamilton's equations (4.9) are $\dot{x} = X_H(x)$. Lecture 5 used the operator $D_f = \{f, \cdot\}$. By antisymmetry, $D_f = -X_f$. We use X_f from now on.

Example 6.2 (Three generators). The following flows solve $\mathrm{d}x/\mathrm{d}s = X_f(x)$. Differentiation verifies each displayed solution and its initial value.

1. *Momentum generates translations.* We have $X_{p_i} = \partial/\partial q_i$, so $\varphi_s^{p_i}$ adds s to q_i and fixes the other coordinates.
2. *Angular momentum generates rotations.* On $\mathbb{R}^6$ let $L_z = q_1 p_2 - q_2 p_1$. By (6.1) and (5.10), $X_{L_z} u = \{u, L_z\} = -u \times e_z = e_z \times u$ for $u = q$ and $u = p$: the flow turns q and p together about e_z, counter-clockwise, through the angle s.
3. *A rotation of the phase plane* ($n = 1$). The function $f = \frac{1}{2}(q^2 + p^2)$ gives $X_f = (p, -q)$ and
$$\varphi_s^f(q, p) = (q\cos s + p \sin s,\ \ p\cos s - q \sin s). \tag{6.2}$$

Sign fact. The flow (6.2) is *clockwise* in the (q, p) plane: it carries $(1, 0)$ to $(\cos s, -\sin s)$ (Figure 6.1). The oscillator of Example 5.16 turns the same way, and the counter-clockwise rotation is generated by $-\frac{1}{2}(q^2 + p^2)$.

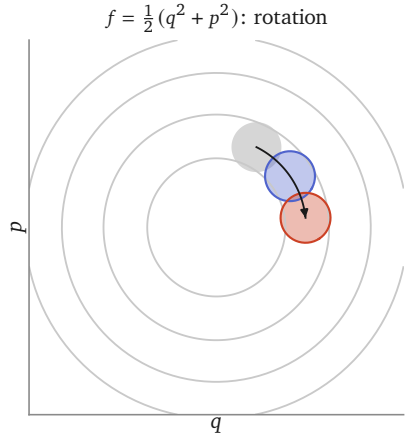


Figure 6.1: The flow of $f = \frac{1}{2}(q^2 + p^2)$: a disc of states (grey) and its images at $s = 0.5$ and $s = 1$, running clockwise along the level sets of f (faint).

Since $J^{\mathsf{T}} = -J$ and $J^{\mathsf{T}} J = I$,

$$\nabla H \cdot X_H = \nabla H^{\mathsf{T}} J \nabla H = 0, \qquad |X_H| = |\nabla H|\,.$$

At a point where $\nabla H \neq 0$, the gradient is normal to the level set through the point and X_H is tangent to it, with the same length. Thus energy is conserved, since $X_H H = \{H, H\} = 0$. The same argument shows that every observable f is constant along its own flow.

6.2 Bracket preservation and the Jacobi identity

In this section $H \in C^\infty(\mathcal{M})$ has a complete flow φ_t (Definition 1.9), and $f_t = f \circ \varphi_t$ is the Heisenberg evolution of Theorem 5.15: $\partial_t f_t = \{f_t, H\}$, and f_t is the only C^1 family with this property and $f_0 = f$.

Definition 6.3 (Poisson automorphism). A C^∞ bijection $\psi\colon \mathcal{M} \to \mathcal{M}$ with C^∞ inverse is a *Poisson automorphism* if $\{f \circ \psi, g \circ \psi\} = \{f, g\} \circ \psi$ for all $f, g \in C^\infty(\mathcal{M})$.

Theorem 6.4 (Theorem B: flows preserve the bracket · *proved modulo T2 (cited)*)**.** *For all $f, g \in C^\infty(\mathcal{M})$ and all $t \in \mathbb{R}$,*

$$\{f, g\} \circ \varphi_t = \{f \circ \varphi_t, g \circ \varphi_t\}. \tag{6.3}$$

Each φ_t is a Poisson automorphism.

Proof. Let $h_t = \{f_t, g_t\}$. By the smoothness assertion of Theorem 5.15, f_t and g_t are C^∞ in (t, x), so h is C^∞ and ∂_t commutes with the x-derivatives in the bracket:

$$\partial_t h_t = \{\partial_t f_t, g_t\} + \{f_t, \partial_t g_t\} = \{\{f_t, H\}, g_t\} + \{f_t, \{g_t, H\}\} = \{\{f_t, g_t\}, H\} = \{h_t, H\},$$

where the third equality is the Jacobi identity in the form (5.7). Also $h_0 = \{f, g\}$. By the uniqueness part of Theorem 5.15, $h_t = \{f, g\} \circ \varphi_t$. Each φ_t is a C^∞ bijection with inverse φ_{-t} (Theorem 1.6). ∎

Taking a bracket commutes with evolving both observables. The proof does not require H to represent energy. Thus *if X_f is complete, every φ_s^f is a Poisson automorphism*, so the translations and rotations of Example 6.2 preserve the bracket. For incomplete flows the same argument, run on the domain W of Theorem 1.6, gives (6.3) wherever φ_t is defined.

For $n = 1$ the bracket of two functions is the Jacobian determinant of the pair: $\{Q, P\} = \det \partial(Q, P)/\partial(q, p)$. With $f = q$ and $g = p$, (6.3) reads $\det D\varphi_t = \{q_t, p_t\} = 1$: *a one-degree-of-freedom Hamiltonian flow preserves area* (Figure 6.2). § 6.6 and Lecture 7 extend this to every n.

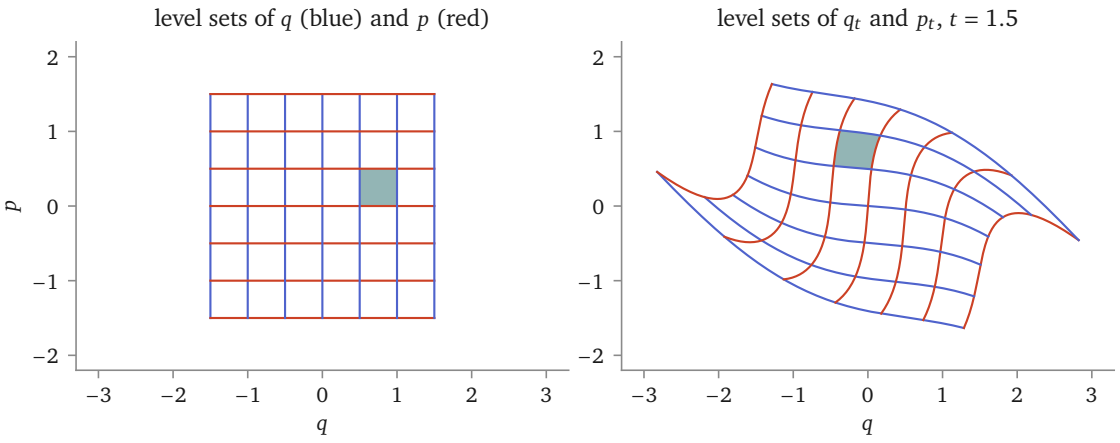


Figure 6.2: Theorem B for $f = q$, $g = p$. Left: the grid of level sets of q and p. Right: the level sets of q_t and p_t for the pendulum, $t = 1.5$, which are the grid lines moved by φ_{-t}. Since $\{q_t, p_t\} = \{q, p\} \circ \varphi_t = 1$, the shaded cell keeps its area $\frac{1}{4}$ (computed from the moved boundary).

Lecture 11 meets brackets whose matrix varies with the point. For those, Jacobi is a real condition, and the next theorem says what it means: the argument above works for other brackets, and it can be reversed. Let $U \subseteq \mathbb{R}^m$ be open, let $\Pi(x)$ be a C^∞ antisymmetric $m \times m$ matrix on U, and let $\{f, g\}_\Pi = \nabla f^{\mathsf{T}} \Pi \nabla g$ (Remark 5.6): bilinear, antisymmetric and Leibniz, as in Proposition 5.3; $m = 2n$ and $\Pi = J$ is our bracket, and nothing below uses that m is even. The Hamiltonian H evolves observables along the flow ψ_t of $\Pi \nabla H$, since $\nabla f \cdot \Pi \nabla H = \{f, H\}_\Pi$.

Theorem 6.5 (Jacobi ⇔ flows preserve the bracket · *proved modulo T1, T2 and smooth cut-offs (cited)*)**.** *The bracket $\{\cdot,\cdot\}_\Pi$ satisfies the Jacobi identity if and only if, for every $H \in C^\infty(U)$ with $\Pi\nabla H$ complete, every map ψ_t of its flow satisfies $\{f\circ\psi_t, g\circ\psi_t\}_\Pi = \{f,g\}_\Pi\circ\psi_t$ for all f, g.*

Proof. ($\Rightarrow$) The proofs of Theorem 5.15 and Theorem 6.4 use the bracket only through $\frac{\mathrm{d}}{\mathrm{d}t} f(\psi_t x) = \{f,H\}_\Pi(\psi_t x)$, through antisymmetry and through Jacobi; they hold verbatim for $\{\cdot,\cdot\}_\Pi$.

($\Leftarrow$) The hypothesis speaks only of complete flows, and an arbitrary H need not have one. To test Jacobi for H near a point, we first change H away from that point so that its flow is complete (Figure 6.3).

Fix f, g, H and $x_0 \in U$. Choose closed balls $B \subset K \subset U$ centred at x_0, and a C^∞ function χ equal to 1 on B and 0 outside K (a smooth cut-off; Lee, *Introduction to Smooth Manifolds*, Ch. 2). The field $F = \Pi\nabla(\chi H)$ vanishes outside K, so it is complete: a solution through a zero of F is constant (Theorem 1.3), so a solution starting in K never reaches a point outside K, and the blow-up alternative applies. Differentiate the hypothesis for χH at $t = 0$; the flow is C^∞ in (t, x) (Theorem 1.6), so ∂_t passes through the gradients, and at every point

$$\{\{f,\chi H\}_\Pi, g\}_\Pi + \{f,\{g,\chi H\}_\Pi\}_\Pi = \{\{f,g\}_\Pi, \chi H\}_\Pi .$$

At x_0 both sides involve derivatives of χH of order at most two, and $\chi H = H$ on B. So the identity holds at x_0 with H in place of χH. It is the second form of (5.7), which is (5.6) by antisymmetry. ∎

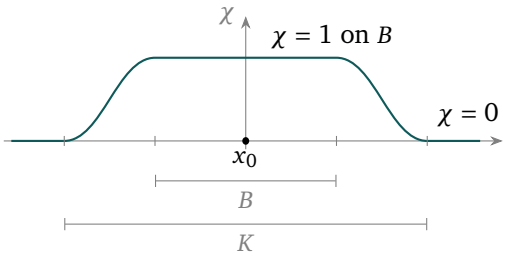


Figure 6.3: The cut-off χ. The function χH equals H on the ball B round x_0 and vanishes outside the larger ball K, so its flow is complete.

The meaning of Jacobi. The Jacobi identity is the statement that time evolution preserves the bracket, infinitesimally.

6.3 The Lie bracket, and $X_{\{f,g\}}$

A vector field $X = (a_1, \ldots, a_m)$ is identified with the first-order operator $X = \sum_k a_k \partial_k$; its components are recovered as $a_k = X x_k$. So X_f is both the vector $J\nabla f$ and the operator $h \mapsto \{h, f\}$.

Definition 6.6 (Lie bracket)**.** The *Lie bracket* of vector fields $X = \sum a_k\partial_k$ and $Y = \sum b_k\partial_k$ is the commutator $[X,Y] = XY - YX$. By Lemma 5.4 it is again a vector field, with components

$$[X,Y]_k = Xb_k - Ya_k, \qquad \text{that is} \qquad [X,Y] = DY\,X - DX\,Y, \tag{6.4}$$

where DX is the Jacobian matrix of the components.

The bracket is bilinear and antisymmetric, and it satisfies the Jacobi identity, since commutators of operators do. The symbol $[\cdot,\cdot]$ also denotes the commutator of matrices. The two differ by a sign on linear fields: for $X(v) = Av$ and $Y(v) = Bv$, (6.4) gives $[X, Y]\,(v) = (BA - AB)v$, the *negative* of the matrix commutator $[A, B]$. § 6.7 uses this. Lecture 12 uses the Lie bracket itself: observables with vanishing bracket have commuting fields.

Theorem 6.7 (Hamiltonian fields and brackets)**.** *For all $f, g \in C^\infty(\mathcal{M})$,*

$$X_{\{f,g\}} = -\left[X_f, X_g\right]. \tag{6.5}$$

Proof. Apply both sides to an observable h. By (6.1),

$$\begin{aligned}\left[X_f, X_g\right] h &= X_f\{h,g\} - X_g\{h,f\} = \{\{h,g\},f\} - \{\{h,f\},g\}\\ &= \{f,\{g,h\}\} + \{g,\{h,f\}\} = -\{h,\{f,g\}\},\end{aligned}$$

by antisymmetry and then the Jacobi identity (5.6). The right side is $-X_{\{f,g\}}h$. Two first-order operators that agree on every h (in particular on the coordinates) are equal. ∎

An example. In one degree of freedom take $f = q^2/2$ and $g = p^2/2$. Then $X_f = -q\,\partial_p$ and $X_g = p\,\partial_q$, and (6.4) gives

$$\left[X_f, X_g\right] = -q\,\partial_q + p\,\partial_p, \qquad \{f,g\} = qp, \qquad X_{qp} = q\,\partial_q - p\,\partial_p, \tag{6.6}$$

so $\left[X_f, X_g\right] = -X_{\{f,g\}}$.

The map $f \mapsto X_f$ is linear and *reverses* brackets; $f \mapsto D_f = -X_f$ preserves them, $D_{\{f,g\}} = \left[D_f, D_g\right]$, which is the first form of (5.7).

Corollary 6.8 (Hamiltonian fields form a Lie algebra)**.** *The Hamiltonian vector fields on $\mathcal{M}$ form a Lie algebra under $[\cdot,\cdot]$. The equality $X_f = 0$ holds if and only if $\nabla f = 0$, that is, f is locally constant. If $\mathcal{M}$ is connected, two observables generate the same field if and only if they differ by a constant.*

Proof. Closure is (6.5). The matrix J is invertible, so $J\nabla f = 0$ exactly when $\nabla f = 0$. Apply this to $f - g$. ∎

6.4 Noether reciprocity

Theorem 6.9 (Noether's theorem, Hamiltonian form · *proved; (ii)⇒(i) and (iii)⇒(i) use T1 (cited)*)**.** *For $H, f \in C^\infty(\mathcal{M})$ the following are equivalent:*

(i) $\{f, H\} = 0$*;*

(ii) *f is conserved: it is constant along every motion of H;*

(iii) *H is invariant under the flow of f: $H(\varphi_s^f x) = H(x)$ wherever $\varphi_s^f x$ is defined.*

Proof. (i)⇔(ii) is Proposition 5.11. For (iii), by (6.1) along the flow of f,

$$\frac{\mathrm{d}}{\mathrm{d}s} H(\varphi_s^f x) = \{H, f\}\,(\varphi_s^f x) = -\{f, H\}\,(\varphi_s^f x).$$

If (i) holds, the derivative vanishes on the whole interval of definition, which contains 0, so (iii) holds. If (iii) holds, the derivative vanishes at $s = 0$, so $\{f, H\}\,(x) = 0$; through every x passes an integral curve (Theorem 1.3). ∎

The statement is symmetric, since exchanging f and H in the vanishing bracket changes only a sign. Thus *f is conserved by the motion of H if and only if H is conserved by the flow of f*.

If the flow of f is complete, it also carries motions of H to motions. Each $\psi = \varphi_s^f$ is a Poisson automorphism (Theorem 6.4) with $H \circ \psi = H$ by invariance, so $X_H(h \circ \psi) = \{h \circ \psi, H \circ \psi\} = (X_H h) \circ \psi$ for every observable h. Let γ be a motion of H. Then $\frac{\mathrm{d}}{\mathrm{d}t} h(\psi(\gamma(t))) = (X_H h)(\psi(\gamma(t)))$ for every h, so $\psi \circ \gamma$ is a motion too. By Corollary 6.8, on a connected $\mathcal{M}$ the symmetry determines the conserved quantity up to a constant.

Point transformations: the cotangent lift

Which observables generate transformations of positions? Let $U \subseteq \mathbb{R}^n$ be open, $\mathcal{M} = U \times \mathbb{R}^n$, and let $\xi \in C^\infty(U, \mathbb{R}^n)$ be a complete vector field on U with flow h^s.

Proposition 6.10 (The cotangent lift · *proved modulo T1, T5 (cited)*)**.** *The flow of the observable $f(q,p) = p \cdot \xi(q)$ is*

$$\varphi_s^f(q,p) = \big(h^s(q),\ Dh^s(q)^{-\mathsf{T}} p\big), \qquad s \in \mathbb{R}. \tag{6.7}$$

Proof. We have $X_f = (\partial f/\partial p, -\partial f/\partial q) = (\xi(q), -D\xi(q)^{\mathsf{T}} p)$. The q-equation $\mathrm{d}q/\mathrm{d}s = \xi(q)$ does not involve p, and is solved by $q(s) = h^s(q_0)$. Let $M(s) = Dh^s(q_0)$; by the variational equation of Theorem 1.8, $M' = D\xi(q(s))\,M$, $M(0) = I$. Then $p(s) = M^{-\mathsf{T}} p_0$ satisfies

$$p' = -M^{-\mathsf{T}} M'^{\mathsf{T}} M^{-\mathsf{T}} p_0 = -D\xi(q(s))^{\mathsf{T}} p(s),$$

the p-equation, with $p(0) = p_0$. By uniqueness (Theorem 1.3) this is the flow, and it is defined for all s. ∎

(6.7) is the covector law (4.3) applied to the point transformation h^s: positions move by h^s, and momenta move as covectors must (Figure 6.4). Translations ($\xi = e$, $f = e \cdot p$) and rotations ($\xi = e \times q$, $f = p \cdot (e \times q) = e \cdot \mathbf{L}$) are of this kind: these are the generators of Example 6.2.

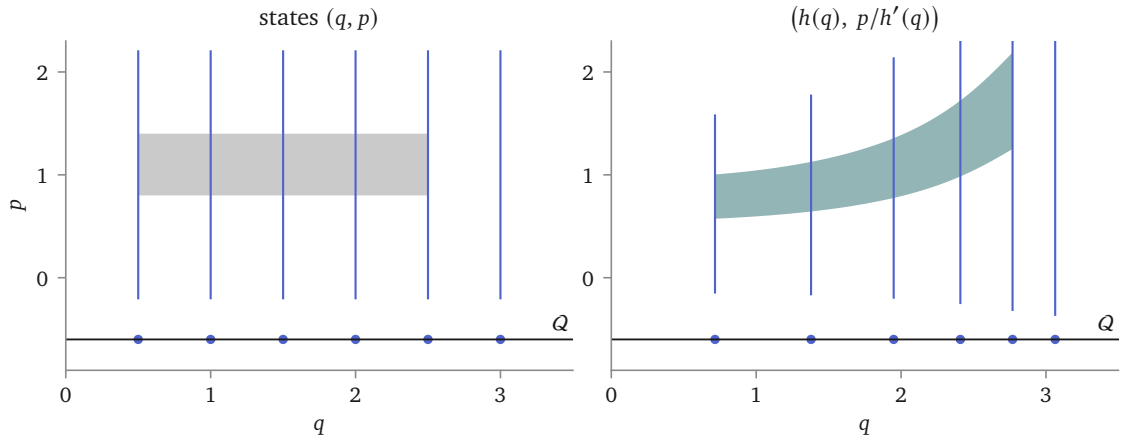


Figure 6.4: A lifted point transformation, $n = 1$: $(q,p) \mapsto (h(q), p/h'(q))$ with $h(q) = q + 0.45 \sin q$. Vertical lines go to vertical lines, and the grey rectangle keeps its area. Below: the configuration line and the points $h(q)$.

For a time-independent strict point symmetry ($G = 0$), the charge of Theorem 3.14 is this $p \cdot \xi$. A quasi-symmetry has the charge $p \cdot \xi - G$ instead. Theorem 6.9 admits every observable f; Lecture 10 meets a conserved one that is quadratic in p.

6.5 The symplectic form

The bracket $\{f, g\} = \nabla f^{\mathsf{T}} J \nabla g$ uses f and g only through their gradients. Removing them leaves a bilinear form on vectors, and that form is what a Jacobian matrix can preserve. The matrix J of (4.9) has $J^{\mathsf{T}} = -J = J^{-1}$, $J^2 = -I$ and $\det J = 1$.

Definition 6.11 (The symplectic form). The *symplectic form* on $\mathbb{R}^{2n}$ is

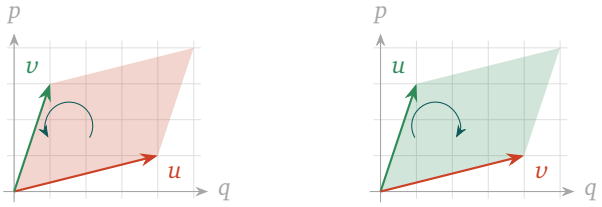
$$\omega(u, v) = u^{\mathsf{T}} J v = \sum_{i=1}^{n} \left(u_{q_i} v_{p_i} - u_{p_i} v_{q_i} \right), \tag{6.8}$$

where $u = (u_q, u_p)$ and $v = (v_q, v_p)$.

It is bilinear; it is *antisymmetric*, $\omega(u, v) = -\omega(v, u)$, because $J^{\mathsf{T}} = -J$; and it is *non-degenerate*: if $\omega(u, v) = 0$ for every v then $J^{\mathsf{T}} u = 0$, and J is invertible, so $u = 0$. Antisymmetry gives $\omega(u, u) = 0$: every vector is ω-orthogonal to itself, which no inner product allows.

What it measures

For $n = 1$, (6.8) is $u_q v_p - u_p v_q = \det \begin{pmatrix} u & v \end{pmatrix}$, the *oriented area* of the parallelogram spanned by u and v, in that order: positive when v lies anticlockwise from u, and changing sign when the two are exchanged (Figure 6.5).



$\omega(u, v) = +2.75$ $\omega(u, v) = -2.75$

Figure 6.5: One degree of freedom: $\omega(u, v)$ is the area of the parallelogram on u and v, signed by the sense in which the boundary u, v is traversed. The right panel is the same pair of vectors with the names exchanged. Here $u = (2, \frac{1}{2})$, $v = (\frac{1}{2}, \frac{3}{2})$ and $\omega(u, v) = 2 \cdot \frac{3}{2} - \frac{1}{2} \cdot \frac{1}{2}$.

For $n > 1$, (6.8) is the *sum of the oriented areas of the projections of that parallelogram onto the n planes* (q_i, p_i); the projections onto (q_1, q_2), (q_1, p_2) and the other coordinate planes do not enter. That is what the notation $\omega = \sum_i \mathrm{d}q_i \wedge \mathrm{d}p_i$ records: one oriented area for each pair (q_i, p_i).

Bracket, field and form are one structure

Proposition 6.12 (ω recovers the bracket). *For all $f, g \in C^\infty(\mathcal{M})$ and all $v \in \mathbb{R}^{2n}$,*

$$\omega(X_f, v) = \nabla f \cdot v, \qquad \text{and} \qquad \omega(X_f, X_g) = \{f, g\}. \tag{6.9}$$

Moreover X_f is the only vector field satisfying the first identity.

Proof. We have $\omega(X_f, v) = (J\nabla f)^{\mathsf{T}} J v = \nabla f^{\mathsf{T}} J^{\mathsf{T}} J v = \nabla f \cdot v$, since $J^{\mathsf{T}} J = I$. Putting $v = X_g = J\nabla g$ gives $\nabla f^{\mathsf{T}} J \nabla g$, which is $\{f, g\}$ by (5.1). If $\omega(w, v) = \nabla f \cdot v$ for all v, then $\omega(w - X_f, v) = 0$ for all v, so $w = X_f$ by non-degeneracy. ∎

Thus the bracket, Hamiltonian vector fields, and the form ω determine one another.

6.6 The symplectic group; flows are symplectic

Definition 6.13 (Symplectic matrix). A real $2n \times 2n$ matrix A is *symplectic* if

$$A^{\mathsf{T}} J A = J.$$

The symplectic matrices form the set $\mathrm{Sp}(2n)$.

Equivalently, $\omega(Au, Av) = \omega(u, v)$ for all $u, v \in \mathbb{R}^{2n}$.

Proposition 6.14 ($\mathrm{Sp}(2n)$ is a group). *If $A \in \mathrm{Sp}(2n)$ then $\det A = \pm 1$; A is invertible with $A^{-1} = -JA^{\mathsf{T}}J \in \mathrm{Sp}(2n)$; and $AB \in \mathrm{Sp}(2n)$ for $B \in \mathrm{Sp}(2n)$. For $n = 1$, $A^{\mathsf{T}} J A = (\det A) J$, so $\mathrm{Sp}(2) = \mathrm{SL}(2, \mathbb{R})$.*

Proof. Taking determinants in $A^{\mathsf{T}} J A = J$ gives $(\det A)^2 \det J = \det J$ and $\det J = 1$, so $(\det A)^2 = 1$. In particular A is invertible. Multiplying $A^{\mathsf{T}} J A = J$ on the left by $-J$ and using $J^2 = -I$ gives $(-JA^{\mathsf{T}}J)A = -J^2 = I$, and a left inverse of an invertible matrix is its inverse. Then $(A^{-1})^{\mathsf{T}} J A^{-1} = (A^{-1})^{\mathsf{T}}(A^{\mathsf{T}} J A) A^{-1} = J$, and $(AB)^{\mathsf{T}} J (AB) = B^{\mathsf{T}} J B = J$. For $n = 1$ expand the 2×2 product. ∎

Symplectic is stronger than volume-preserving

For $n = 1$, Proposition 6.14 says that symplectic means $\det A = 1$: area and orientation preserved, nothing more. For $n \geq 2$ it is strictly stronger. With e_{q_i}, e_{p_i} the standard basis vectors, take, in the order (q_1, q_2, p_1, p_2),

$$A = \mathrm{diag}\big(2, 1, 1, \frac{1}{2}\big), \qquad \det A = 1. \tag{6.10}$$

Then $\omega(Ae_{q_1}, Ae_{p_1}) = 2$ and $\omega(Ae_{q_2}, Ae_{p_2}) = \frac{1}{2}$, instead of 1 and 1: the map doubles one projected area and halves the other. The two errors cancel in the determinant and not in ω.

Theorem 6.15 (Hamiltonian flows are symplectic · *proved modulo T2, T5 (cited)*). *Let $H \in C^\infty(\mathcal{M})$ with flow φ_t. Then $D\varphi_t(x) \in \mathrm{Sp}(2n)$ for every x and every t for which $\varphi_t x$ is defined.*

Proof. Fix x and let $M(t) = D\varphi_t(x)$, which is C^1 in t on the open interval $I_x \ni 0$ where the motion exists (Theorem 1.6). Differentiating $\partial_t \varphi_t x = X_H(\varphi_t x)$ in x, the variational equation of Theorem 1.8 reads

$$M'(t) = J \operatorname{Hess} H(\varphi_t x)\, M(t), \qquad M(0) = I, \tag{6.11}$$

since $DX_H = D(J\nabla H) = J \operatorname{Hess} H$. Write $S = \operatorname{Hess} H(\varphi_t x)$, which is *symmetric*. Then

$$\frac{\mathrm{d}}{\mathrm{d}t}\big(M^{\mathsf T} J M\big) = M^{\mathsf T}\big[(JS)^{\mathsf T} J + J(JS)\big]M = M^{\mathsf T}\big[SJ^{\mathsf T}J + J^2 S\big]M = M^{\mathsf T}\big[S - S\big]M = 0.$$

So $M(t)^{\mathsf T} J M(t) = M(0)^{\mathsf T} J M(0) = J$ on I_x. ∎

Only two facts were used: that X_H is J times a gradient, and that a Hessian is symmetric. For $n = 1$ this is the area statement of § 6.2; Lecture 7 proves in every dimension that volume is preserved.

6.7 Quadratic Hamiltonians

Quadratic Hamiltonians are the systems whose flows are linear, and they give the finite-dimensional Lie algebra of § 5.7. For a symmetric $2n \times 2n$ matrix S put

$$H_S(x) = \frac{1}{2} x^{\mathsf T} S x, \qquad \text{so that} \qquad X_{H_S}(x) = J\nabla H_S = JSx.$$

Every homogeneous quadratic is an H_S for exactly one symmetric S, and the motion is the linear equation $\dot x = JSx$, solved by $x(t) = e^{tJS}x_0$ for all $t \in \mathbb{R}$.

Proposition 6.16 (Hamiltonian matrices). *For a real $2n \times 2n$ matrix X the following are equivalent:*

(a) $X^{\mathsf T}J + JX = 0$;

(b) $X = JS$ *with* $S = -JX$ *symmetric;*

(c) $e^{tX} \in \mathrm{Sp}(2n)$ *for all* $t \in \mathbb{R}$.

These matrices form $\mathfrak{sp}(2n)$, a linear space of dimension $n(2n+1)$.

Proof. (b)⇒(a): $(JS)^{\mathsf T}J + J^2 S = SJ^{\mathsf T}J - S = 0$. (a)⇒(b): put $S = -JX$; then $S^{\mathsf T} = X^{\mathsf T}J = -JX = S$ and $JS = X$. (a)⇔(c): $Z(t) = e^{tX^{\mathsf T}} J e^{tX}$ has $Z' = e^{tX^{\mathsf T}}(X^{\mathsf T}J + JX)e^{tX}$ and $Z(0) = J$, so (a) gives $Z \equiv J$, and differentiating $Z \equiv J$ at $t = 0$ gives (a). Since $S \mapsto JS$ is a bijection, the dimension is that of the symmetric matrices. ∎

Theorem 6.17 (Quadratics and matrices). *For symmetric S, T,*

$$\{H_S, H_T\} = H_{S'}, \qquad S' = SJT - TJS, \qquad JS' = [JS, JT]\,, \tag{6.12}$$

the commutator of matrices. Hence $\mathfrak{sp}(2n)$ is closed under $[\cdot,\cdot]$, and $JS \mapsto H_S$ is a linear isomorphism of $\mathfrak{sp}(2n)$ onto the homogeneous quadratics that carries the matrix commutator to the Poisson bracket.

Proof. We have $\{H_S, H_T\} = (Sx)^{\mathsf T} J (Tx) = x^{\mathsf T} S J T x$. A quadratic form depends only on the symmetric part of its matrix, and $(SJT)^{\mathsf T} = -TJS$, so $x^{\mathsf T}SJTx = \tfrac12 x^{\mathsf T} S' x = H_{S'}$ with S' as stated; S' is symmetric. Multiplying S' by J, $JS' = JSJT - JTJS = [JS, JT]$. Closure of $\mathfrak{sp}(2n)$ follows, since S' is symmetric. ∎

The signs agree with § 6.3: the bracket of linear fields is the negative of the matrix commutator, so $X_{\{H_S,H_T\}} = [JS, JT]\,x = -\left[X_{H_S}, X_{H_T}\right]$, which is (6.5).

The oscillator with $m\nu = 1$ is the first example. For $H = \frac{1}{2}\nu(q^2 + p^2)$, $S = \nu I$ and $JS = \nu J$, whose exponential $e^{t\nu J}$ is the clockwise rotation through the angle νt of (6.2) and Figure 6.1: an element of $\mathrm{Sp}(2)$ for every t, as the exponential criterion of Proposition 6.16 requires.

The computation (6.6) gives an example with a nonzero bracket. The functions $f = q^2/2$ and $g = p^2/2$ are H_S and H_T with

$$S = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad T = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \qquad JS = \begin{pmatrix} 0 & 0 \\ -1 & 0 \end{pmatrix}, \quad JT = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Their commutator is

$$[JS, JT] = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = JS', \qquad S' = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

and $H_{S'} = qp = \{f, g\}$, as Theorem 6.17 states.

Summary.

- *Handed on:* every observable generates a flow, $X_f = J\nabla f$ (Definition 6.1); complete flows preserve the bracket, and a bracket satisfies Jacobi exactly when its flows do (Theorems 6.4 and 6.5); $X_{\{f,g\}} = -\left[X_f, X_g\right]$ (Theorem 6.7); f is conserved exactly when H is invariant under the flow of f (Theorem 6.9), and point symmetries are cotangent lifts (Proposition 6.10).
- *The form behind the bracket:* $\omega(u, v) = u^{\mathsf{T}} J v$, with $\omega(X_f, X_g) = \{f, g\}$ (Proposition 6.12); flows have symplectic Jacobians (Theorem 6.15); the quadratic Hamiltonians are $\mathfrak{sp}(2n)$ (Theorem 6.17).
- *Next:* Lecture 7 puts the states beside the observables: volume is preserved, densities are carried along, and the second answer is complete.

References

Marsden and Ratiu, *Introduction to Mechanics and Symmetry*, §2.7 and §13.1. Takhtajan, *Quantum Mechanics for Mathematicians*, §2.7. Arnold, *Mathematical Methods of Classical Mechanics*, §§39–41. Meyer and Offin, *Introduction to Hamiltonian Dynamical Systems and the N-Body Problem*, §§2.1–2.2 and p. 196.